

Broader Impact

ToT is a framework that empowers LMs to more autonomously and intelligently make decisions and solve problems. While current tasks are limited to reasoning and search problems, future applications involving interaction with external environments or humans could bring potential danger, e.g. facilitating harmful uses of LMs. On the other hand, ToT also improves the interpretability of model decisions and the opportunity for human alignment, as the resulting representations are readable, high-level language reasoning instead of implicit, low-level token values.

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References

- [1] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- [2] C. Browne, E. J. Powley, D. Whitehouse, S. M. M. Lucas, P. I. Cowling, P. Rohlfshagen, S. Tavener, D. P. Liebana, S. Samothrakis, and S. Colton. A survey of monte carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in Games*, 4:1–43, 2012.
- [3] M. Campbell, A. J. Hoane Jr, and F.-h. Hsu. Deep blue. *Artificial intelligence*, 134(1-2):57–83, 2002.
- [4] X. Chen, M. Lin, N. Schärli, and D. Zhou. Teaching large language models to self-debug, 2023.
- [5] A. Chowdhery, S. Narang, J. Devlin, M. Bosma, G. Mishra, A. Roberts, P. Barham, H. W. Chung, C. Sutton, S. Gehrmann, et al. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.
- [6] A. Creswell and M. Shanahan. Faithful reasoning using large language models. *arXiv preprint arXiv:2208.14271*, 2022.
- [7] N. D. Daw, Y. Niv, and P. Dayan. Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature neuroscience*, 8(12):1704–1711, 2005.
- [8] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, and G. Neubig. Pal: Program-aided language models, 2023.
- [9] S. Hao, Y. Gu, H. Ma, J. J. Hong, Z. Wang, D. Z. Wang, and Z. Hu. Reasoning with language model is planning with world model. *arXiv preprint arXiv:2305.14992*, 2023.
- [10] P. E. Hart, N. J. Nilsson, and B. Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968. doi: 10.1109/TSSC.1968.300136.
- [11] P. E. Hart, N. J. Nilsson, and B. Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.
- [12] W. Huang, P. Abbeel, D. Pathak, and I. Mordatch. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents, 2022.
- [13] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, et al. Inner monologue: Embodied reasoning through planning with language models. *arXiv preprint arXiv:2207.05608*, 2022.

- [14] J. Jung, L. Qin, S. Welleck, F. Brahman, C. Bhagavatula, R. L. Bras, and Y. Choi. Maieutic prompting: Logically consistent reasoning with recursive explanations. *arXiv preprint arXiv:2205.11822*, 2022.
- [15] D. Kahneman. *Thinking, fast and slow*. Macmillan, 2011.
- [16] D. Kahneman, S. Frederick, et al. Representativeness revisited: Attribute substitution in intuitive judgment. *Heuristics and biases: The psychology of intuitive judgment*, 49(49-81):74, 2002.
- [17] G. Kim, P. Baldi, and S. McAleer. Language models can solve computer tasks, 2023.
- [18] B. Liu, Y. Jiang, X. Zhang, Q. Liu, S. Zhang, J. Biswas, and P. Stone. Llm+p: Empowering large language models with optimal planning proficiency, 2023.
- [19] X. Lu, S. Welleck, P. West, L. Jiang, J. Kasai, D. Khashabi, R. L. Bras, L. Qin, Y. Yu, R. Zellers, N. A. Smith, and Y. Choi. Neurologic a*esque decoding: Constrained text generation with lookahead heuristics. In *North American Chapter of the Association for Computational Linguistics*, 2021.
- [20] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegrefe, U. Alon, N. Dziri, S. Prabhmoey, Y. Yang, S. Welleck, B. P. Majumder, S. Gupta, A. Yazdanbakhsh, and P. Clark. Self-refine: Iterative refinement with self-feedback, 2023.
- [21] A. Newell, J. C. Shaw, and H. A. Simon. Report on a general problem solving program. In *IFIP congress*, volume 256, page 64. Pittsburgh, PA, 1959.
- [22] A. Newell, H. A. Simon, et al. *Human problem solving*. Prentice-Hall, 1972.
- [23] OpenAI. Gpt-4 technical report. *ArXiv*, abs/2303.08774, 2023.
- [24] D. Paul, M. Ismayilzada, M. Peyrard, B. Borges, A. Bosselut, R. West, and B. Faltings. Refiner: Reasoning feedback on intermediate representations, 2023.
- [25] A. Radford, K. Narasimhan, T. Salimans, I. Sutskever, et al. Improving language understanding by generative pre-training. *OpenAI blog*, 2018.
- [26] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [27] I. Schlag, S. Sukhbaatar, A. Celikyilmaz, W. tau Yih, J. Weston, J. Schmidhuber, and X. Li. Large language model programs, 2023.
- [28] N. Shinn, B. Labash, and A. Gopinath. Reflexion: an autonomous agent with dynamic memory and self-reflection, 2023.
- [29] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, et al. Mastering the game of go without human knowledge. *nature*, 550(7676):354–359, 2017.
- [30] S. A. Sloman. The empirical case for two systems of reasoning. *Psychological bulletin*, 119(1): 3, 1996.
- [31] K. E. Stanovich. *Who is rational? Studies of individual differences in reasoning*. Psychology Press, 1999.
- [32] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- [33] S. Verma, J. Fu, S. Yang, and S. Levine. Chai: A chatbot ai for task-oriented dialogue with offline reinforcement learning. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4471–4491, 2022.

- [34] E. Wallace, N. Tomlin, A. Xu, K. Yang, E. Pathak, M. Ginsberg, and D. Klein. Automated crossword solving. *arXiv preprint arXiv:2205.09665*, 2022.
- [35] L. Wang, W. Xu, Y. Lan, Z. Hu, Y. Lan, R. K.-W. Lee, and E.-P. Lim. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models, 2023.
- [36] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- [37] Z. Wang, S. Cai, A. Liu, X. Ma, and Y. Liang. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents, 2023.
- [38] J. Wei, X. Wang, D. Schuurmans, M. Bosma, E. Chi, Q. Le, and D. Zhou. Chain of thought prompting elicits reasoning in large language models. *arXiv preprint arXiv:2201.11903*, 2022.
- [39] Y. Xie, K. Kawaguchi, Y. Zhao, X. Zhao, M.-Y. Kan, J. He, and Q. Xie. Decomposition enhances reasoning via self-evaluation guided decoding, 2023.
- [40] S. Yang, O. Nachum, Y. Du, J. Wei, P. Abbeel, and D. Schuurmans. Foundation models for decision making: Problems, methods, and opportunities, 2023.
- [41] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.
- [42] S. Zhang, Z. Chen, Y. Shen, M. Ding, J. B. Tenenbaum, and C. Gan. Planning with large language models for code generation. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=Lr8c00tYbfL>.
- [43] D. Zhou, N. Schärli, L. Hou, J. Wei, N. Scales, X. Wang, D. Schuurmans, C. Cui, O. Bousquet, Q. Le, et al. Least-to-most prompting enables complex reasoning in large language models. *arXiv preprint arXiv:2205.10625*, 2022.
- [44] X. Zhu, J. Wang, L. Zhang, Y. Zhang, R. Gan, J. Zhang, and Y. Yang. Solving math word problem via cooperative reasoning induced language models. *arXiv preprint arXiv:2210.16257*, 2022.